Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/11/2023
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Gap Junctions and Microgravity Induced Bone Loss		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	hjdonahue@vcu.edu	**Fax:**	FY 804-827-0290
PI Organization Type:	UNIVERSITY	**Phone:**	804-828-7263
Organization Name:	Virginia Commonwealth University		
PI Address 1:	Department of Biomedical Engineering		
PI Address 2:	401 West Main Street		
PI Web Page:			
City:	Richmond	**State:**	VA
Zip Code:	23284-3067	**Congressional District:**	3
Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/24/2018	**End Date:**	12/31/2023
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
Contact Email:	annamaria.j.ruby@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per PI and NSSC information (Ed., 4/11/23) NOTE: End date changed to 12/31/2022 per NSSC information. (Ed. 8/25/22) NOTE: End date changed to 07/23/2022 per NSSC information. (Ed. 11/30/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang, Yue Ph.D. (Virginia Commonwealth University)		
Grant/Contract No.:	80NSSC18K1473		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts exposed to extended periods of unloading experience significant bone loss. As NASA undertakes an exploration initiative involving long-duration spaceflight and exploration of asteroids and Mars, astronauts will encounter prolonged periods of unloading followed by subsequent extra-vehicular activities in a gravity environment. Loss of bone mass and strength will complicate these activities and impair mission efficiency. For instance, considering the magnitude of bone loss occurring during spaceflight (0.5-1.5% loss of bone mineral density (BMD)/month), a fall on an already weakened skeletal system could result in fracture. Therefore, development of countermeasures for microgravity-induced bone loss is critical. We previously demonstrated that osteoblast- and osteocyte-specific deficiency in the gap junctional protein connexin 43 (Cx43) is protective against bone loss induced by unloading via hindlimb suspension (HLS). However, the mechanism by which this occurs is unknown. Osteocytes are well situated in bone to detect and communicate mechanical signals to effector cells and therefore play a pivotal role in bone's ability to sense and respond to unloading from exposure to microgravity. Therefore, we will examine the hypothesis that Cx43 deficiency attenuates the response of osteocytes to unloading resulting from simulated microgravity. Additionally, we will examine whether the response of osteocytes exposed to simulated microgravity in vitro can predict bone response to simulated microgravity in vivo. Using an in vivo and in vitro model, we will complete the following two aims. Aim 1: Examine the response of wild-type and Cx43-deficient (via CRISPR gene editing) murine osteocytes (OCY454) to simulated microgravity in vitro using a High Aspect Ratio Vessel (HARV). Aim 2: Examine the effect of simulated microgravity in vivo on bones from osteocyte-specific Cx43-deficient mice. Cx43 will be selectively deleted from mice using a Cre recombinase approach similar to what we previously used to delete Cx43 from osteoblasts and osteocytes. Using well-established experimental paradigms like the HARV apparatus and HLS greatly increases the probability of successfully completing our aims and attaining important, novel findings. Collectively, our infrastructure and expertise, in combination with multiple unique reagents, place us in an unparalleled position to rapidly and significantly advance knowledge pertaining to the mechanisms underlying microgravity-induced bone loss. These aims and the experiments designed therein represent a substantive departure from our current understanding and will reveal novel targets for countermeasures.
Rationale for HRP Directed Research:	
	The proposed research will establish the mechanism underlying the protective effect of Cx43 against microgravity-induced bone loss. This will help fill gaps in our understanding of gravity-sensing mechanisms. We will also gain insight as to whether in vitro models of microgravity, such as culturing cells in a HARV apparatus, induce similar changes in osteocytic gene expression as do in vivo models such as HLS. From the perspective of microgravity countermeasures, this information will guide the choice of potential therapeutics. On the ground, concomitant bone loss is frequently experienced by individuals confined to bed for long durations, whether as a result of neurological injury, trauma, or neuromuscular disease (e.g., cerebral palsy and muscular dystrophy). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of bone occurs. In the case of recoverable disease or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Understanding the mechanisms involved in bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population.
Research Impact/Earth Benefits:	The proposed research is innovative because it provides the first simultaneous examination of effects of simulated microgravity on genetically modified osteocytes in vivo and in vitro. A few studies have examined the effects of simulated microgravity on bone in vivo and in vitro,but no study has compared the effects of simulated microgravity on osteocytes or genetically modified mice in vivo and in vitro in the same study. Osteocytes play a key role in bone’s mechanosensitivity and adaptation to changes in mechanical loading. Furthermore, we are investigating the novel concept that Cx43 deficiency protects against microgravity-induced bone loss through a non-canonical mechanism involving an interaction between Cx43 and the WNT/ß-catenin pathway. Finally, we will use innovative transcriptomic approaches to compare the response of primary osteocytes exposed to simulated microgravity in vivo to osteocytic OCY454 cells exposed to simulated microgravity in vitro. In this proposal we are leveraging our experience and expertise in bone biology (Donahue) and gene expression (Buck). Our team brings forward a collaborative and unique skill set in order to ensure a complete and comprehensive investigation of the response of osteocytic cells to simulated microgravity.
	Progress on Aim 1. As detailed in previous annual reports, we made considerable progress on aim 1. In addressing aim 1, which focused on the response of wild-type (WT) and Cx43-deficient (via CRISPR gene editing) murine osteocytes (OCY454) to simulated microgravity in vitro, WT Ocy454 osteocytic cells and Cx43 deficient (Cx43 KO) cells generated by CRISPR were seeded onto Alvetex 12 well scaffold inserts (500,000 cells/insert; Reprocell) and cultured for 10 days at 37°C in aMEM (10% FBS; 2% Pen/Strep). At 10 days, scaffolds were placed in rotary cell culture vessels (RCCS; Synthecon) to simulate three conditions: 1) microgravity (Sim (MG)); 2) rotary ground (Rot Grd); or 3) static ground (Stat Grd) conditions, as previously documented, for 72 hours. Relative expression of Sim MG and Rot Grd genes (compared to Stat Grd) for osteocytic apoptosis (DDIT3) and resorption (RANKL,OPG) were assessed by the micro-CT method. Soluble sclerostin expression in the media was assessed by ELISA (R&D Systems) and normalized to scaffold cell number by PicoGreen DNA Assay (Invitrogen). Data (n = 4-8 RCCS vessels per group) were analyzed by 2-way ANOVA with Tukey’s post-hoc test ($p<0.05$). Seventy-two hours of Sim MG significantly increased relative DDIT3 expression in WT but not Cx43 KO osteocytes. RANKL/OPG expression was not affected by Sim MG or Cx43 deficiency. Sclerostin protein secretion was significantly upregulated by Sim MG in WT osteocytes but not Cx43 KO osteocytes. These results partially support our hypothesis and suggest that Cx43 deficiency protects against unloading induced osteocyte apoptosis and sclerostin secretion, a potent WNT/B-catenin inhibitor. The RANKL/OPG expression ratio was unaffected by Sim MG and Cx43 KO despite previous reports. This may be due to experimental differences between the studies and our own, such as osteocyte cell type used, MLO-Y4 versus Ocy454, or differences in unloading conditions. Furthermore, despite increased apoptotic signaling in WT osteocytes undergoing SIM MG versus Cx43 KO, cell counts were not statistically different at seventy-two hours between groups. This suggests that increased cell death due to apoptotic signaling may not be apparent in WT osteocytes till later timepoints or be compensated by enhanced proliferation.

Task Progress:	These results suggest that Cx43 deficiency reduces apoptotic gene expression and sclerostin expression in response to simulated microgravity. Thus, the protective effects of osteoblast and osteocyte Cx43 deficiency on unloading induced bone loss may be to preserve osteocyte viability and boost WNT/B-catenin signaling. Progress on Aim 2. For aim 2, we were able to examine mice with constitutive deletion of Cx43. Mice with constitutive deletion of Cx43 in osteocytes only (Dmp1 Cre) display only a mild bone loss phenotype making them a good choice for our experiments. Additionally, we were able to expand the study and examine muscle changes in these mice. DMP1 Lineage Cx43 Deficient mice underwent hind limb unloading (HLU) to simulate microgravity or normal weight-bearing activity (controls) for 3 weeks. Data was analyzed for effect of treatment (HLU, control) and genotype (WT, Cx43 cKO) by 2-WAY repeated measures ANOVA with Tukey's multiple comparisons test (*p< 0.05). HLU and Cx43 deficiency synergistically decreased normalized gastrocnemius but not soleus mass (only affected by HLU) (p<0.05). HLU overall decreased tetanic torque following plantar flexion compared to controls by -20% (p<0.05). In contrast, Cx43 deficiency independently increased tetanic torque by +26% compared to controls, abrogating much of the disuse effect (p<0.05). In uninjured diaphyseal femurs, HLU and Cx43 deficiency synergistically decreased cortical and metaphyseal bone volume (p<0.05; ~2-3 fold relative to controls). Although fracture calluses trended toward decreased callus volume with HLU (-22% vs. controls) and reduced mineralization with Cx43 (-16% vs WT) independently, these effects didn't reach significance. Our results suggest that Cx43 deficiency in DMP1 lineage cells may protect against impairments in muscle function with HLU but not changes in bone mass or fracture healing, at least in female mice. We also had the ability to expand the study and examine the effect of HLU on fracture healing. Mice with constitutive deletion of Cx43 in osteocytes only (Dmp1 Cre) underwent HLU for 3 weeks following skeletal maturity (26 weeks) to simulate extended microgravity exposure. Then, mice had their right femur fractured by open surgical dissection (stabilized with 24-gauge pin) and were allowed to heal for 14 days. Mice given normal cage activity throughout the experiment served as weight-bearing controls (GC). Mouse injured and uninjured contralateral hindlimb bone and muscle properties were analyzed for the effects of treatment (HLU, control) and genotype (WT, Cx43 cKO) by 2-WAY ANOVA with Tukey's multiple comparisons test (*p< 0.05). Cx43 deficient bones demonstrated significantly reduced cortical area fraction at baseline compared to WT mice. With HLU, both WT and CX43 deficient mice showed significant diaphyseal bone loss and gastrocnemius muscle mass (p <0.05); however, this effect wasn't dependent on genotype. Looking at fractured femurs, there were trends of differential effects between treatment and genotype. For example, WT mice with HLU showed decreased callus bone volume and bone volume fraction compared to ground control with CX43 deficient calluses showing opposite effects. In addition, Cx43 deficient mice under ground control conditions showed decreased callus formation compared to WT mice. These results suggest that Cx43 deficiency in DMP1 lineage cells may protect against impairments in fracture healing with HLU, and therefore microgravity. We have continued with development of a tamoxifen inducible osteocyte specific Cx43 deficient mouse model using DMP1 Cre-ERT2 construct. Importantly, DMP1 Cre-ERT2 shows only a small degree of leakiness (~10-20% of osteocytes) and minimal muscle targeting. This novel model will allow us to more accurately determine the role of postnatal Cx43 expression on osteocyte apoptosis and senescence in the context of disuse without a confounding basal phenotype. We plan to generate inducible, osteocyte specific Cx43 deficient mice. Inducible, osteocyte specific Cx43 deficient mice will be generated by crossing the 10kb-Dmp1-Cre-ERT2 transgenic line D77 (DMP1 Cre-ERT2; Jax #029594) with Cx43 floxed (Cx43fl/fl; Jax Stock #008039) mice. Furthermore, the Cre reporter Ai14 allele (Rosa-CAG-LSL-tdTomato Jax Stock #007914) will be introduced into our experimental mouse line to visualize and quantify DMP1 Cre-ERT2 activated cells. Mice will be given tamoxifen for 3 consecutive days (3X; 75mg/kg; i.p.) 1 week before unloading to induce recombination. DMP1 Cre-ERT2 Cx43fl/fl Ai14 mice given tamoxifen will serve as osteocyte Cx43 deficient mice whereas Cx43fl/fl Ai14 littermate mice given tamoxifen will serve as controls. Thus far, we have successfully bred the Cx43 and Ai14 alleles to homozygosity in a C57BL/6J congenic mouse strain and crossed to the DMP1 Cre-ERT2 line. Experimental mice are aging for experiments starting in June 2023.

Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Buettmann EG, Goldscheitter GM, Hoppock GA, Friedman MA, Suva LJ, Donahue HJ. "Similarities between disuse and age-induced bone loss." J Bone Miner Res. 2022 Jun 30;37(8):1417-34. Review. https://doi.org/10.1002/jbmr.4643 ; PMID: 35773785; PMCID: PMC9378610 , Jun-2022
Articles in Peer-reviewed Journals	Boyan BD, Berger MB, Nelson FR, Donahue HJ, Schwartz Z. "The biological basis for surface-dependent regulation of osteogenesis and implant osseointegration." J Am Acad Orthop Surg. 2022 Apr 5. http://dx.doi.org/10.5435/JAAOS-D-21-00523 ; PMID: 35383608; PMCID: PMC9464448 , Apr-2022
Articles in Peer-reviewed Journals	Friedman MA, Abood A, Senwar B, Zhang Y, Maroni CR, Ferguson VL, Farber CR, Donahue HJ. "Genetic variability affects the skeletal response to immobilization in founder strains of the diversity outbred mouse population." Bone Rep. 2021 Dec;15:101140. https://doi.org/10.1016/j.bonr.2021.101140 ; PMID: 34761080; PMCID: PMC8566767 , Dec-2021
Articles in Peer-reviewed Journals	Juhl OJ 4th, Buettmann EG, Friedman MA, DeNapoli RC, Hoppock GA, Donahue HJ. "Update on the effects of microgravity on the musculoskeletal system." npj Microgravity. 2021 Jul 23;7(1):28. Review. https://doi.org/10.1038/s41526-021-00158-4 ; PMID: 34301942; PMCID: PMC8302614 , Jul-2021